

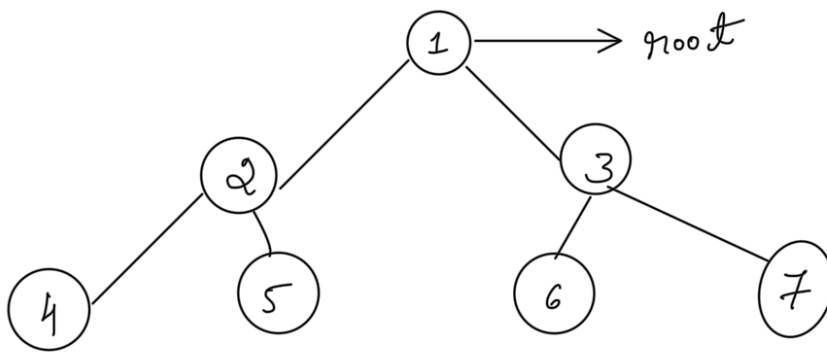
Trees

DAY 171
28/Dec/2025

Level Order Traversal

Level order Traversal is defined as follows:

- Visit the root
- While traversing level l , keep all the elements at $l+1$ in queue
- Go to the next level and visit all the nodes at the level
- Repeat this until all levels are completed.



level 1

level 2

level 3

```
static void levelOrder (TreeNode root) {
```

```
    if (root == null) return;
```

```
    Queue<TreeNode> q = new LinkedList<>();
```

```
    q.add(root);
```

```
    while (!q.isEmpty()) {
```

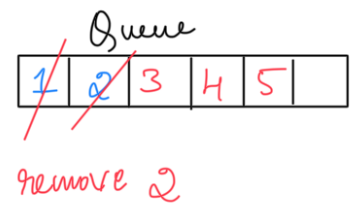
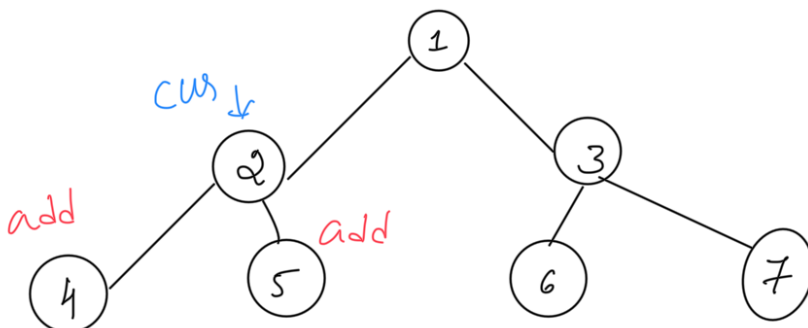
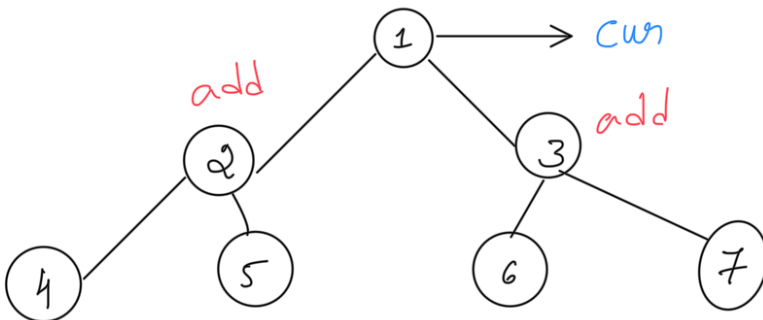
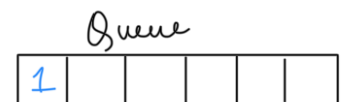
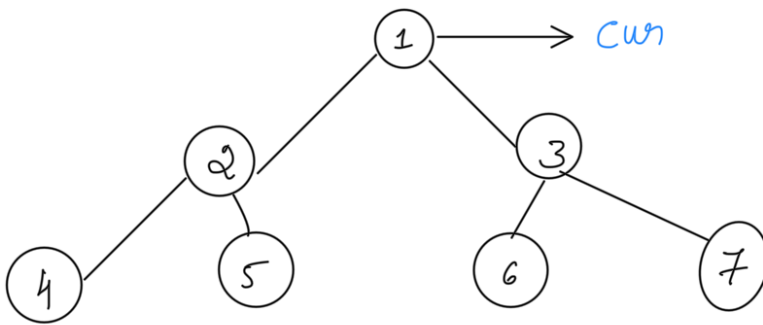
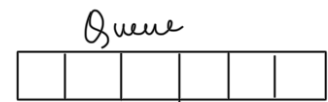
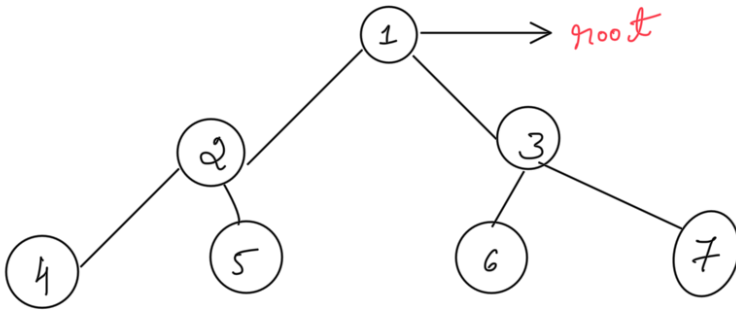
```
        TreeNode curr = q.poll();
```

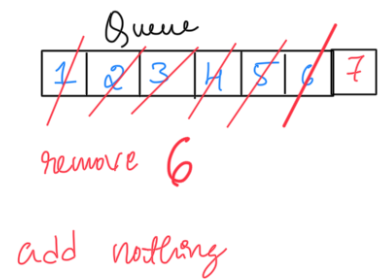
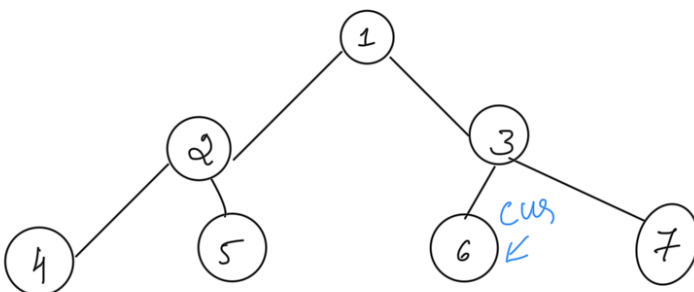
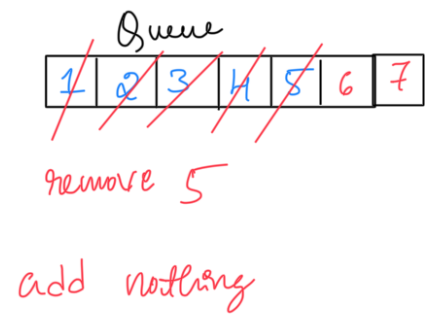
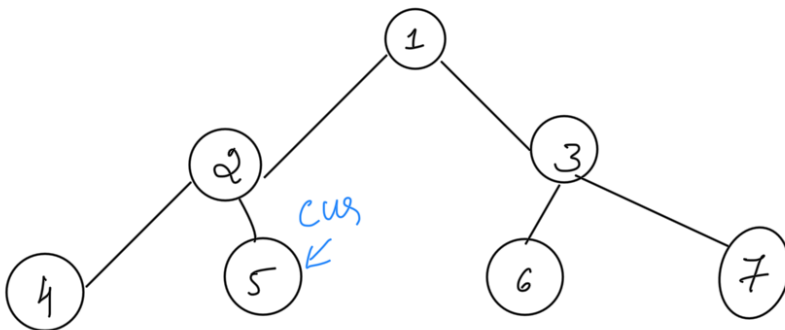
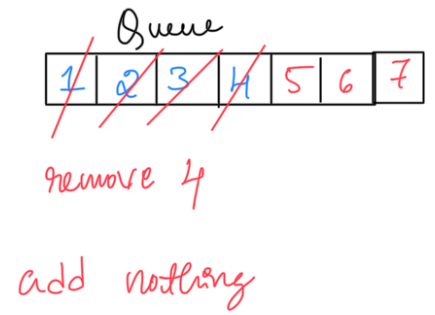
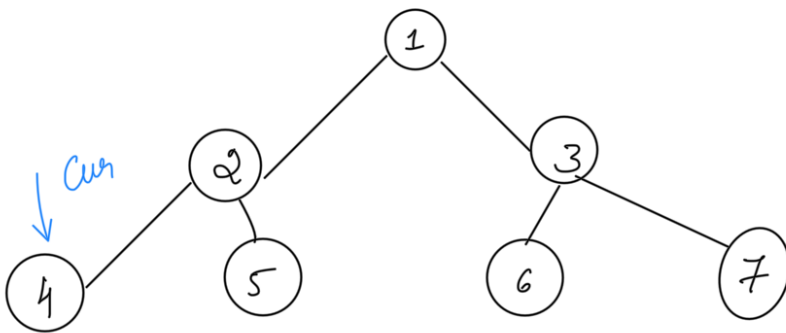
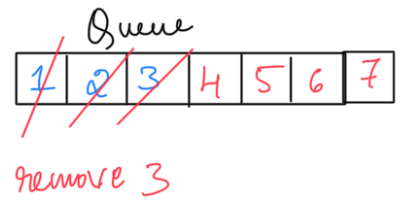
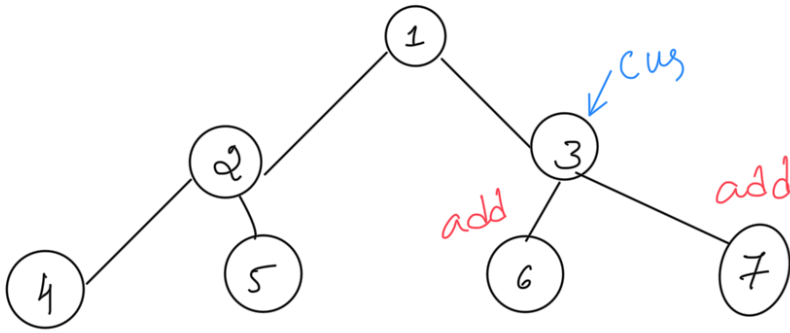
```
        System.out.print(curr.data + " →");
```

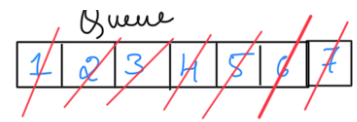
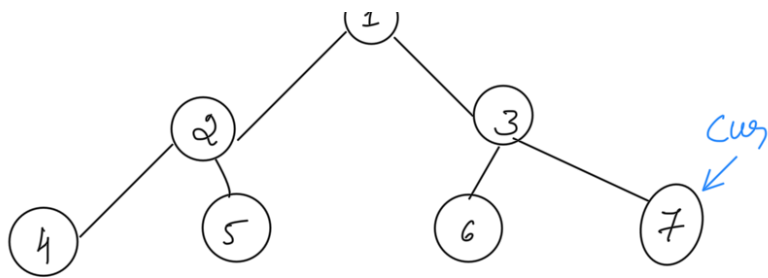
```
        if (curr.left != null) q.add(curr.left);
```

```
        if (curr.right != null) q.add(curr.right);
```

```
    }
```







remove 7

add nothing

level order traversal

1	2	3	4	5	6	7
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Time Complexity $O(N)$

Space Complexity $O(n)$

Since, in the worst case, all the nodes on the entire last level could be in the Queue simultaneously